

CariSurg MedTech Pathways Research Project

Project Title: Development of an Interpretable Digital Triage System for Dynamic Risk Assessment in Emergency Care

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Statement of the Problem:

Emergency department overcrowding delays care for critically ill patients because traditional triage relies on static, point-in-time assessments that fail to capture real-time physiological deterioration in the waiting room. Current AI triage solutions operate as uninterpretable "black boxes," hindering clinical adoption and alignment with established hospital protocols. This 12-week pilot introduces a transparent digital triaging model that translates real-time patient data into actionable risk categories using recognized NEWS2 and SIRS frameworks, safely accelerating clinical throughput and mitigating under-triage errors.

Previous Work:

Extensive peer-reviewed literature establishes that artificial intelligence can significantly optimise emergency department workflows. ^[18] conducted a systemised scoping review across EMBASE, MEDLINE, and Web of Science and found that machine learning models consistently demonstrated superior discrimination and predictive accuracy compared to traditional triage systems, successfully enhancing disease identification, predicting hospital admission needs, and improving resource allocation. Critically, the review also flagged that

AI algorithms risk producing overconfident outputs, which poses direct patient safety concerns and underscores the need for interpretable rather than purely high-performing models. ^[17] reinforced this multimodal direction, finding through a narrative synthesis of 20 peer-reviewed papers that combining free-text nurse notes with structured vital sign data consistently improved predictive performance over structured data alone, with NLP models achieving high accuracy in predicting triage scores, hospital admissions, and critical illness. ^[12] further demonstrated this through the TriageIntelli framework, confirming that fusing heterogeneous clinical data streams yields superior predictive accuracy compared to single-modality baselines, though they noted that heavy data dependency and variable performance across settings remain significant implementation barriers.

The interpretability of these systems has emerged as a distinct and critical research focus. ^[13] conducted a systematic PRISMA-guided review of ten empirical studies measuring the direct impact of explainable AI on clinician trust and found that XAI can significantly increase trust compared to black-box models, but only when explanations are concise, clear, and relevant to actual clinical workflows. Importantly, their analysis revealed that complex or contradictory explanations can actively undermine trust or cause clinicians to accept flawed advice, proving that the design of interpretability is as consequential as its presence. This finding directly informs the design decision in this project to map outputs onto NEWS2 and SIRS frameworks rather than presenting raw algorithmic scores. ^[20] demonstrated the practical viability of this approach, developing an interpretable machine learning triage tool using 323,509 ED episodes in Singapore that achieved an AUROC of 0.823 for 30-day mortality prediction while outperforming all six established clinical benchmarks including NEWS and MEWS, confirming that interpretability and clinical-grade accuracy are not mutually exclusive goals. However, significant gaps

persist in the existing literature. ^[16] identified the most critical structural limitation, noting that the majority of current AI triage models evaluate patients using a single snapshot taken at arrival and completely ignore physiological deterioration during prolonged waiting room stays. ^[8] quantified the cost of this limitation directly, demonstrating that dynamic early warning scores incorporating temporal vital sign trends outperformed static NEWS2 in predicting ICU admission or death within 24 hours (AUROC 0.906 versus 0.857) and urgent deterioration within four hours (AUROC 0.877 versus 0.829), catching more critical events with the same rate of false alarms. ^[6] identified the second persistent gap, finding through a narrative review of 26 peer-reviewed studies that even high-performing AI triage systems consistently fail to reach clinical deployment because uninterpretable outputs fuel clinician scepticism and prevent alignment with standardised hospital protocols. Two critical gaps therefore emerge from this body of literature and directly shape the design of this pilot. The first is the near-universal reliance on static, single-point triage assessment, with the majority of current models capturing one snapshot at arrival and failing entirely to monitor physiological trajectory during waiting room stays ^{[8][16]}. The second is the persistent failure of even high-performing AI triage systems to reach clinical deployment, driven by uninterpretable outputs that prevent alignment with standardised hospital protocols and erode clinician trust ^{[13][6]}. This pilot directly addresses both gaps.

Literature Review:

Arnaud E, Elbattah M, Gignon M, Dequen G. Deep Learning to Predict Hospitalization at Triage: Integration of Structured Data and Unstructured Text. In: 2020 IEEE International Conference on Big Data (Big Data). IEEE; 2020:4836-4841.
doi:10.1109/BigData50022.2020.9378073

Clinical/Engineering Problem: International Emergency Department (ED) overcrowding causes severe operational issues, directly leading to negative care outcomes such as prolonged patient lengths of stay.

Method Used: A multi-input Deep Learning framework trained on a large-scale dataset of approximately 260,000 ED records from Amiens-Picardy University Hospital in France. The system uses a dual-stream architecture to process mixed data types simultaneously: a standard Multi-Layer Perceptron (MLP) runs numeric and categorical features, while a Convolutional Neural Network (CNN) runs unstructured textual notes in parallel to predict whether a patient will be discharged or hospitalized.

Reported Outcome: The parallel, mixed-data classification model successfully predicted patient hospitalization needs at the triage stage, achieving high predictive accuracy with an area under the receiver operating characteristic curve (ROC-AUC) of approximately 0.83.

Stated Limitation: The abstract is restricted to evaluating retrospective data from a single hospital network (Amiens-Picardy University Hospital) and does not outline any live clinical deployment, prospective evaluation, or methods for resolving model explainability for frontline medical staff.

Tyler S, Olis M, Aust N, et al. Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review. *Cureus*. Published online May 8, 2024. doi:10.7759/cureus.59906

Clinical/Engineering Problem: Emergency department (ED) triage lacks precision and certainty regarding its overall impact when using traditional systems. There is a critical need to improve earlier diagnosis, patient risk prioritization, resource allocation (like predicting length of stay), and determining which patients accurately require hospitalization.

Method Used: A systemized literature review was conducted in September 2023 across the EMBASE, Ovid MEDLINE, and Web of Science databases.

Reported Outcome: Machine learning models consistently showed superior discrimination and predictive accuracy compared to traditional triage systems. The algorithms successfully enhanced disease identification, accurately determined when urgent patients needed hospital admission, and improved hospital resource allocation and overall quality of care.

Stated Limitation: The text notes that there is still little known overall about the use of AI in ED triage, and it highlights a specific danger where AI algorithms can give "overconfident answers" that could pose safety risks to patients.

Da'Costa A, Teke J, Origbo JE, Osonuga A, Egbon E, Olawade DB. AI-driven triage in emergency departments: A review of benefits, challenges, and future directions. *International Journal of Medical Informatics*. 2025;197:105838. doi:10.1016/j.ijmedinf.2025.105838

Clinical/Engineering Problem: Emergency Departments (EDs) face severe operational pressure from overcrowding, resource constraints, and traditional triage systems that rely on subjective assessments, leading to a lack of consistency in patient prioritization during peak hours or mass casualty events.

Method Used: A narrative review was conducted analyzing peer-reviewed articles published between 2015 and 2024. The papers were identified through targeted searches across

PubMed, Scopus, IEEE Xplore, and Google Scholar databases, with the findings synthesized into a comprehensive overview.

Reported Outcome: The synthesized findings showed substantial benefits for AI-driven triage, including improved patient prioritization, reduced wait times, and optimized hospital resource allocation, demonstrating a strong potential to transform ED operations and improve patient outcomes in high-pressure environments.

Stated Limitation: Widespread adoption is significantly blocked by operational and technical challenges, specifically data quality issues, algorithmic bias, a lack of clinician trust, and unresolved ethical concerns regarding equitable implementation.

Araouchi Z, Adda M. TriageIntelli: AI-Assisted Multimodal Triage System for Health Centers. *Procedia Computer Science*. 2024;251:430-437. doi:10.1016/j.procs.2024.11.130

Clinical/Engineering Problem: Traditional emergency department triage methods—like the Emergency Severity Index (ESI) and the Canadian Triage and Acuity Scale (CTAS)—suffer from human subjectivity, cannot handle severe overcrowding efficiently, and lead to poor hospital resource allocation.

Method Used: A literature review examining and comparing the evolutionary transition of emergency triage systems from conventional, manual protocols over to modern AI-assisted predictive models.

Reported Outcome: The review found that AI models offer enhanced diagnostic precision, optimized patient prioritization, and a reduction in human error, which collectively streamlines busy emergency room workflows.

Stated Limitation: The implementation of these models faces critical bottlenecks, specifically heavy data dependency, unresolved ethical challenges, and inconsistent, variable performance when deployed across different healthcare settings.

Sitthiprawiat P, Wittayachamnankul B, Sirikul W, Laohavisudhi K. Development and internal validation of an AI-based emergency triage model for predicting critical outcomes in emergency department. *Sci Rep*. 2025;15(1):31212. doi:10.1038/s41598-025-17180-1

Clinical/Engineering Problem: Severe emergency department (ED) overcrowding delays patient care and worsens clinical outcomes because traditional triage systems have inherent limitations in accuracy and consistency when identifying who needs urgent resources.

Method Used: A retrospective analysis of 163,452 ED visits (from 2018 to 2022) compared logistic regression, random forest, and XGBoost machine learning models against the traditional Canadian Triage and Acuity Scale (CTAS). The models evaluated mode of arrival, patient age, and free-text chief complaints using advanced multilingual sentence embeddings.

Reported Outcome: The XGBoost model achieved superior predictive performance for ICU admission and resource needs over traditional CTAS (AUROC 0.917 vs. 0.882).

Incorporating unstructured text data via machine learning significantly enhanced triage accuracy and resource allocation by identifying critically ill patients more effectively.

Stated Limitation: The study relied on a retrospective analysis of historical data from a single facility (Maharaj Nakhon Chiang Mai Hospital), meaning the model was validated internally and its performance may vary when deployed in a live, real-time clinical environment.

Stewart J, Lu J, Goudie A, et al. Applications of natural language processing at emergency department triage: A narrative review. Reddy S, ed. PLoS ONE. 2023;18(12):e0279953. doi:10.1371/journal.pone.0279953

Clinical/Engineering Problem: Standard emergency triage risk stratification techniques struggle to predict clinical outcomes like hospital admission and critical illness because they completely ignore the rich, unstructured human language contained within free-text nursing notes.

Method Used: A scoping review and narrative synthesis screened 3,730 studies across MEDLINE, Embase, Cochrane, Web of Science, and Scopus to evaluate 20 peer-reviewed papers that applied NLP to free-text data acquired at ED triage. The researchers used the Prediction model Risk of Bias Assessment Tool (PROBAST) to systematically evaluate the quality and bias of the included predictive models.

Reported Outcome: NLP models achieved high accuracy in predicting triage scores, hospital admissions, and critical illness. Furthermore, combining free-text data with structured data consistently improved model predictive performance compared to using structured data alone.

Stated Limitation: The findings are heavily restricted because 80% of the reviewed studies carry a high risk of bias, almost all the research is retrospective, and there is a critical shortage of prospective clinical trials or real-world model deployment in live practice.

Rosenbacke R, Melhus Å, McKee M, Stuckler D. How Explainable Artificial Intelligence Can Increase or Decrease Clinicians' Trust in AI Applications in Health Care: Systematic Review. JMIR AI. 2024;3:e53207. doi:10.2196/53207

Clinical/Engineering Problem: The "black box" nature of clinical artificial intelligence creates severe user barriers. If clinicians cannot understand an algorithm's reasoning, they may completely reject its advice. Conversely, if they trust it blindly, they may defer to incorrect machine outputs that conflict with their own medical judgment, leading to dangerous clinical errors.

Method Used: A systematic review was conducted following the PRISMA guidelines, searching the PubMed and Web of Science databases. The researchers screened 778 articles to identify 10 empirical studies that measured the direct impact of Explainable AI (XAI) on clinicians' trust using specific cognition- or affect-based trust metrics. Each included paper was systematically assessed for methodological risk of bias.

Reported Outcome: The review confirmed that XAI can significantly increase clinician trust compared to "black box" AI, but only if the generated explanations are concise, clear, and relevant to actual clinical workflows. It also revealed that providing explanations does not automatically improve trust, proving that trust can be carefully modulated and balanced through smart XAI system design.

Stated Limitation: All ten evaluated empirical studies carried a moderate to high risk of bias. Furthermore, the existing literature relies heavily on abstract, cognitive definitions of trust rather than real-world behavioral measures, and the review notes that complex or contradictory explanations can actively undermine clinical trust or cause clinicians to accept flawed advice.

Xie F, Ong MEH, Liew JNMH, et al. Development and Assessment of an Interpretable Machine Learning Triage Tool for Estimating Mortality After Emergency Admissions. JAMA Netw Open. 2021;4(8):e2118467. doi:10.1001/jamanetworkopen.2021.18467

Clinical/Engineering Problem: Emergency department triage requires highly complex clinical judgment. While existing scoring tools are meant to assist with risk stratification, they have significant built-in limitations and struggle to provide early, highly accurate estimates of a patient's risk of mortality using a small, practical set of variables.

Method Used: A single-site, retrospective cohort study of 323,509 total ED episodes (split into training, validation, and testing cohorts) for patients subsequently admitted to a tertiary hospital in Singapore. The researchers used an interpretable machine learning framework to develop the Score for Emergency Risk Prediction (SERP) tool based on a stripped-down (parsimonious) list of triage variables, comparing its predictive accuracy for 2-, 7-, and 30-day mortality against six established clinical benchmarks (including NEWS and MEWS).

Reported Outcome: The machine learning-derived SERP tool significantly outperformed all traditional clinical benchmark scores across all target timelines, achieving strong predictive power (such as an AUROC of 0.823 for 30-day mortality) while maintaining easy implementation and simple data collection in a fast-paced ED.

Stated Limitation: The tool was derived and tested using a retrospective cohort study from a single hospital site, meaning its real-world clinical performance still needs to be widely applied and validated across different types of healthcare settings and live clinical environments.

Gonem S, Taylor A, Figueredo G, et al. Dynamic early warning scores for predicting clinical deterioration in patients with respiratory disease. Respir Res. 2022;23(1):203. doi:10.1186/s12931-022-02130-6

Clinical/Engineering Problem: The standard National Early Warning Score 2 (NEWS-2) framework fails to account for fine-grained clinical details or how a patient's vital signs change over time (temporal trends), which limits its accuracy when trying to predict sudden health decline or urgent treatment needs.

Method Used: Data-driven time-series features from electronic records across 31,590 respiratory in-patient episodes (from 2015 to 2020 at a large UK hospital trust) were extracted. Researchers manually annotated 1,100 of these episodes to pinpoint the exact timing of clinical deterioration and fed these trend variables into logistic regression models to build a Dynamic Early Warning Score (DEWS), testing its predictive accuracy against standard NEWS-2.

Reported Outcome: The dynamic model (DEWS) significantly outperformed standard NEWS-2 in predicting intensive care unit admission or death within 24 hours (AUROC 0.906 vs. 0.857) and urgent clinical deterioration within 4 hours (AUROC 0.877 vs. 0.829). DEWS caught more critical events ahead of time while maintaining the same low rate of false alarms.

Stated Limitation: The study relied entirely on historical data from respiratory patients at a single acute NHS Trust. The authors explicitly stated that prospective validation studies are still required to see if these dynamic scores actually reduce missed deteriorations and false alarms in live, real-world clinical environments.

Boulitsakis Logothetis S, Green D, Holland M, Al Moubayed N. Predicting acute clinical deterioration with interpretable machine learning to support emergency care decision making. Sci Rep. 2023;13(1):13563. doi:10.1038/s41598-023-40661-0

Clinical/Engineering Problem: Increasing operational pressures in fast-paced Emergency Departments create an urgent need to efficiently identify patients at risk of rapid health decline. Traditional tracking frameworks like NEWS2 can cause alert fatigue or entirely miss high-risk patients who present with low initial scores.

Method Used: A retrospective study analyzing a dataset of 118,886 unplanned hospital admissions at Salford Royal Hospital, UK. The researchers systematically compared machine learning algorithms (logistic regression, gradient-boosted decision trees, and support vector machines) using a patient's first recorded vital signs, lab results, and observations to predict mortality or critical care needs within 24 hours. The team used Shapley Additive exPlanations (SHAP) to map out feature weights and justify model outputs.

Reported Outcome: The machine learning models significantly outperformed the traditional NEWS2 framework, yielding up to a 0.366 increase in average precision, a notable reduction in daily false alarms (alert fatigue), and a drop in biased predictions across different age and sex demographics.

Stated Limitation: The models were built on historical, cross-sectional data from a single hospital site, and the authors explicitly noted that further work is still needed to trial and test these models in actual, live clinical practice.

What it Proves: This study directly evaluates machine learning algorithms (such as gradient-boosted decision trees) against the traditional NEWS2 framework using Electronic Health Record (EHR) data from ED patients.

Liu N, Xie F, Siddiqui FJ, et al. Leveraging Large-Scale Electronic Health Records and Interpretable Machine Learning for Clinical Decision Making at the Emergency Department:

Clinical/Engineering Problem: Growing global demand and overcrowding in Emergency Departments lead to longer waiting times. While machine learning tools can build triage and risk prediction models to help, the "black box" nature of these advanced models limits their actual clinical use and interpretation by staff.

Method Used: A retrospective, single-center study protocol analyzing a massive longitudinal dataset of approximately 1.8 million ED visits (representing over 810,000 unique patients) from a tertiary hospital in Singapore. The researchers used a machine learning tool called AutoScore to create three dynamic, interpretable scores (called SERT) to evaluate patients at different times: on arrival, during their ED stay, and at admission.

Reported Outcome: The system successfully creates a unique scoring system that is both dynamic and transparent. It establishes a clear standard for using large-scale Electronic Health Records (EHRs) and interpretable machine learning to predict severe outcomes, like the need for an intensive care unit (ICU) or inpatient death.

Stated Limitation: The paper describes a retrospective study protocol and system design rather than a finished prospective clinical trial, meaning the models still require further real-world validation to prove their clinical safety and impact in live practice.

What it Proves: The authors outline the clinical imperative for moving away from 5-level static tools (like the Emergency Severity Index) and standard scores because they are subjective, static, and fail to track patients whose conditions rapidly change or deteriorate during an ED stay.

Chang YH, Lin YC, Huang FW, et al. Using machine learning and natural language processing in triage for prediction of clinical disposition in the emergency department. BMC Emerg Med. 2024;24(1):237. doi:10.1186/s12873-024-01152-1

Clinical/Engineering Problem: Accurate triage is essential to allocate hospital resources efficiently and reduce overall patient stay times. However, manual triage decisions are frequently subjective and vary significantly from one medical provider to another, leading to dangerous errors where patients are either over-triaged or under-triaged.

Method Used: A multi-center retrospective study using a large-scale dataset of 213,984 total non-trauma adult ED visits across two separate institutions (China Medical University Hospital for internal training and Asia University Hospital for external validation). The researchers combined structured data with unstructured free-text nursing notes processed via Natural Language Processing (NLP) to train six distinct machine learning algorithms. They directly compared the models' ability to predict patient hospital disposition using actual emergency physicians' (EPs) manual clinical predictions as a baseline reference.

Reported Outcome: Every machine learning model significantly outperformed human emergency physicians in accurately predicting critical care admissions, general ward admissions, or ED deaths. A fine-tuned Random Forest model demonstrated the most robust

predictive performance, and further data analysis explicitly proved that incorporating unstructured text notes via NLP substantially boosted the algorithm's accuracy over using structured data alone.

Stated Limitation: The study relied entirely on historical data collected between 2018 and 2019 across two specific hospital facilities. Although the model underwent successful external validation, it required precise post-modeling threshold adjustments to balance clinical sensitivity against positive predictive value before it could be deemed safely applicable for real-life triage environments.

Okada Y, Ning Y, Ong MEH. Explainable artificial intelligence in emergency medicine: an overview. Clin Exp Emerg Med. 2023;10(4):354-362. doi:10.15441/ceem.23.145

Clinical/Engineering Problem: Clinicians naturally lack deep artificial intelligence expertise, leading them to perceive modern machine learning triage tools as a "black box." This creates severe trust and adoption issues in high-stakes emergency environments.

Method Used: A comprehensive literature review mapping out the core definitions, requirements, and categories of Explainable AI (XAI) specifically within emergency medicine. The framework breaks explainability down into three structural tiers: pre-modeling explainability, inherently interpretable models, and post-modeling explainability techniques (such as text justification, simplification, and feature relevance mapping).

Reported Outcome: The review demonstrates that incorporating XAI into emergency care fulfills four critical clinical needs: justification of machine decisions, system control, performance improvement, and medical discovery. Using specific post-modeling tools like text justification allows complex algorithms to communicate clearly with clinical staff.

Stated Limitation: Transitioning these models into live clinical settings faces steep implementation challenges. The paper highlights that current success is bottlenecked by a historical lack of close, cross-disciplinary collaboration between frontline clinicians, system developers, and medical researchers.

Boonstra A, Laven M. Influence of artificial intelligence on the work design of emergency department clinicians a systematic literature review. BMC Health Serv Res. 2022;22(1):669. doi:10.1186/s12913-022-08070-7

Clinical/Engineering Problem: Artificial Intelligence remains a new, unfamiliar technology to many emergency care professionals. This lack of familiarity with its actual capabilities creates a knowledge gap regarding how these tools alter the day-to-day work design and decision-making processes of frontline clinicians.

Method Used: A systematic literature review conducted in accordance with the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines. The researchers screened and analyzed 34 peer-reviewed papers published over a five-year period that specifically focused on the implementation and clinical utility of AI within the Emergency Department.

Reported Outcome: The review found that AI applications are primarily deployed at the initial triage stage to establish patient care trajectories, improve the clinical decision-making process, and help relieve overcrowding. The literature demonstrates that while AI cannot replace human clinicians, it significantly optimizes work design by providing reliable decision support that lightens the increasing clinical burden.

Stated Limitation: The integration of AI in emergency care is still in its nascent stages, and current technology lacks the scope or capability to outperform overall human clinical skills. Furthermore, the review relies on a high-level synthesis of existing literature rather than evaluating a specific, newly deployed algorithmic model or prospective clinical trial.

De Freitas L, Goodacre S, O'Hara R, Thokala P, Hariharan S. Qualitative exploration of patient flow in a Caribbean emergency department. *BMJ Open*. 2020;10(12):e041422. doi:10.1136/bmjopen-2020-041422

Clinical/Engineering Problem: Emergency departments are highly complex systems where poor patient flow stalls care. While developed countries have researched this heavily using quantitative numbers, developing regions like the Caribbean are severely under-represented in medical literature, leaving local hospitals without clear insights into the unique organizational bottlenecks slowing down their emergency care.

Method Used: A qualitative, multi-method study conducted at a single major tertiary-level emergency department in Trinidad and Tobago. The researchers directly tracked patient and staff journeys using a single observer who gathered data through non-participant observations, informal conversational interviews, and observational process mapping to build a visual workflow. The data was then analyzed using thematic analysis.

Reported Outcome: The study successfully mapped out six key organizational categories that dictate patient flow: work processes, layout design, material resources, nursing staff levels, non-clinical staff, and external departments. It created a conceptual framework showing that while built-in strategies like patient streaming and front-loading medical investigations help speed up care, staff frequently have to rely on intuitive actions to bypass remaining system bottlenecks.

Stated Limitation: The study relied entirely on a single observer collecting data at a single hospital site. Because of this, the findings are highly context-specific to that particular institution, meaning the results must be validated in other healthcare settings before they can be widely applied across the region.

Olakotan OO, Mohd Yusof M. The appropriateness of clinical decision support systems alerts in supporting clinical workflows: A systematic review. *Health Informatics J*. 2021;27(2):14604582211007536. doi:10.1177/14604582211007536

Clinical/Engineering Problem: Clinical Decision Support Systems (CDSS) generate an unsustainably high volume of inappropriate, unoptimized alerts that constantly interrupt medical workflows. This triggers severe alert fatigue, leading clinicians to routinely silence,

disable, or completely ignore system notifications, which ultimately undermines patient safety.

Method Used: A systematic literature review analyzing 76 papers (comprising retrospective cohort, qualitative, quantitative, and mixed-method studies) published between 1997 and 2018. The researchers searched seven major electronic databases (including PubMed, Scopus, and IEEE) and used a combined approach incorporating a socio-technical framework, the "five rights of CDSS," and Lean principles to categorize the root factors affecting alert appropriateness.

Reported Outcome: The review mapped out the specific technological, human, organizational, and process factors that dictate alert efficiency. It conclusively proved that the primary reason for high alert override rates in live hospital practice is that the vast majority of CDSS alerts were fundamentally not built using established human factor engineering methods and design principles.

Stated Limitation: The study is a high-level systematic literature review restricted to historical papers published up to 2018. While it provides clear structural recommendations and insights for future interface design, it does not build, deploy, or clinically validate a specific algorithmic solution or live alerting mechanism in an active emergency department.

Proposed Solution:

This project proposes a four-component digital triage decision support system designed for real-time deployment within the Mercer General Hospital Emergency Department. Each component directly addresses a specific gap identified in the literature, transitioning from the static, single-input models that dominate current research towards a transparent, continuously updating framework built around clinical workflows staff already use.

Component 1: Multi-Source Ingestion Engine

The system ingests patient data through two simultaneous streams rather than a single data entry point. The first stream pulls structured numerical vitals directly from the triage desk monitor, capturing heart rate, blood pressure, respiratory rate, temperature, GCS, and oxygen saturation in real time. The second stream runs a lightweight natural language processing scanner across free-text nurse notes, automatically flagging high-risk clinical language such as "crushing chest pain" or "diaphoretic" to contribute to the risk score even when objective vitals appear borderline. ^[5] validated this dual-input architecture across 213,984 ED visits at two institutions, demonstrating that every machine learning model tested significantly outperformed human emergency physicians in predicting critical care admissions, and that incorporating unstructured nurse note text via NLP substantially boosted accuracy over structured data alone. ^[17] corroborated this finding across 20 peer-reviewed papers, confirming that combining free-text and structured data consistently outperforms either stream in isolation.

Component 2: Time-Delta Trajectory Tracking

To eliminate the static snapshot limitation identified by ^[16] and ^[8], the system incorporates an

automated re-triage clock. If a patient has been waiting longer than 30 minutes without a vital sign update, the dashboard issues a visual prompt to clinical staff. When updated vitals are entered, the system calculates the directional change from arrival values rather than evaluating the new reading in isolation. A respiratory rate that has climbed from 16 to 24 breaths per minute over one hour triggers an immediate priority upgrade, flagging trajectory rather than threshold alone. ^[8] demonstrated that this temporal approach outperformed static NEWS2 in predicting urgent deterioration within four hours with an AUROC of 0.877 compared to 0.829, catching more critical events at the same false alarm rate. ^[9] further validated the feasibility of dynamic scoring at scale, developing a continuously updated interpretable scoring protocol across 1.8 million ED visits that evaluated patients at arrival, during their stay, and at admission.

Component 3: Glass-Box Logic Layer

To address the interpretability barrier comprehensively identified by ^[6] and ^[13], the system never presents a risk output without an accompanying plain-language clinical explanation. Rather than displaying a raw probability score such as "Risk: 85%", the interface displays the classification mapped directly onto a familiar framework: "High Risk (NEWS2 Score: 6) driven by Respiratory Rate of 25 and Pulse of 110." Every explanation uses clinical terminology staff already recognise rather than algorithmic language they do not. ^[13] established that XAI increases clinician trust only when explanations are concise, clinically relevant, and directly actionable, and that poorly designed explanations can actively reduce trust or cause clinicians to accept flawed outputs. ^[10] identified that post-modelling explainability techniques, particularly text justification and feature relevance mapping, are

the most effective mechanisms for communicating complex algorithmic reasoning to clinicians in high-stakes emergency environments. ^[4] validated SHAP-based feature weighting as the appropriate underlying mechanism, achieving a 0.366 increase in average precision over static NEWS2 while maintaining full model auditability across age and sex demographics.

Component 4: Parallel Shadow Workflow Pilot

Over the 12-week pilot window, the system operates entirely in shadow mode alongside Mercer General's existing manual triage processes. Nurses conduct triage exactly as they currently do. The digital system runs silently in parallel, generating its own NEWS2 and SIRS-aligned risk classifications without displaying them to clinical staff or influencing patient care decisions. A research coordinator logs instances where the system's real-time assessment diverged from the manual triage decision, recording the time differential between the system's escalation flag and the point at which clinical staff identified the same deterioration manually. This produces the primary outcome measure for the pilot: the mean time advantage of the automated system over manual re-triage in identifying patients who deteriorated in the waiting room. Taylor et al. (2025) demonstrated in a randomised controlled trial that variability in nurse agreement with AI recommendations makes shadow-mode deployment the safest implementation approach, positioning the technology as a complement to clinical judgement rather than a replacement. This design choice also directly addresses the cross-disciplinary collaboration gap identified by ^[10] ensuring frontline clinicians, system developers, and researchers are working from the same workflow data before any live deployment decision is made.

Methodology

The workflow diagram below was adapted directly from the validated process map structure published by ^[7], the only qualitative observational study of patient flow conducted in a Caribbean ED to date. That study mapped 143 patient journeys across 203 hours of direct observation at a public teaching hospital ED in Trinidad and Tobago, and identified six categories of organisational factors that facilitate or hinder flow: work processes, design and layout, material resources, nursing staffing, non-clinical staff, and external departments ^[7]. Building the Mercer General workflow on this validated structure, rather than a textbook ED flowchart, ensures the diagram reflects how Caribbean EDs actually operate rather than how Western literature assumes they operate.

Workflow diagram

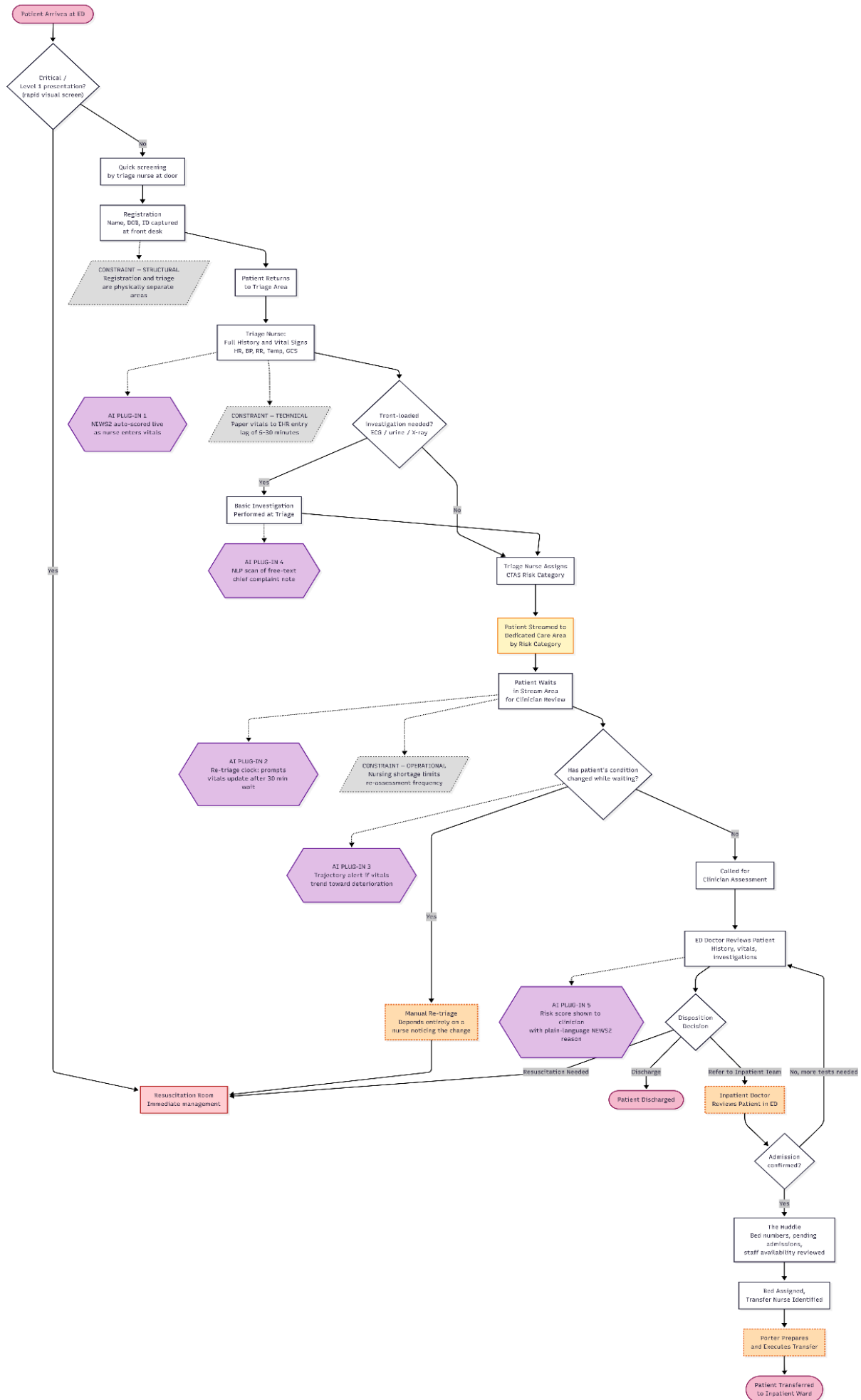


Figure 1. Show the flow diagram tracing the patient journey from arrival to ward transfer using the same structural backbone as ^[7]'s main process map (streaming and triage, front-loaded investigations, the transfer "huddle"), with five AI plug-in points overlaid in purple and three workflow constraints overlaid in grey.

See triage workflow diagram in plots/ on

<https://github.com/baldiea75-tech/carisurg-portfolio.git>

The AI plug-in

AI Plug-in 1: Live NEWS2 scoring at triage. As the triage nurse enters vitals, the system calculates a NEWS2 score in real time rather than requiring a separate manual calculation step.

AI Plug-in 2: Automated re-triage clock. If a patient has waited more than 30 minutes in the stream area without a vitals update, the system issues a visual prompt. This directly targets the static snapshot problem: AI decision support tools that update only once at admission show strong predictive performance on retrospective data but are evaluated using a single point-in-time input, leaving deterioration during the wait undetected ^[19].

AI Plug-in 3: Trajectory alert. Rather than scoring a single reading in isolation, the system flags a worsening trend across consecutive readings, similar in principle to the 10-minute interval re-evaluation approach validated by ^[19], who found that updating predictions throughout a patient's ED stay rather than once at arrival substantially improved the clinical relevance of an admission-prediction model across 154,347 ED visits.

AI Plug-in 4: NLP scan of free-text complaints. A lightweight natural language scan of the chief complaint note flags high-risk language. This plug-in is justified by ^[2], who demonstrated that integrating structured vitals with unstructured free-text triage notes improved hospitalisation prediction over structured data alone.

AI Plug-in 5: Plain-language risk explanation at clinician review. Rather than displaying a raw score, the system maps the output onto NEWS2 language clinicians already use. This design choice is grounded in evidence that clinician trust in AI output depends heavily on how the recommendation is communicated, not only on model accuracy ^[12].

Workflow Constraints

Constraint 1: Physical separation of registration and triage. ^[7] found that the physical separation of registration and triage areas created additional waiting and activity steps that did not exist in the clinical process itself, purely as a consequence of department layout. Any AI system assuming a single continuous data-capture point at "arrival" must account for this gap, since vitals are not captured until the patient physically reaches the second area.

Constraint 2: Nursing staff shortages limiting re-assessment capacity. ^[7] documented that EDs in Caribbean hospitals frequently operate below the nurse staffing levels required for the streaming model to function as designed, forcing nurses to multitask across multiple care streams simultaneously.^[11] similarly found in a systematic review of AI's influence on clinical work design that AI tools succeed only when they reduce rather than add to existing workload, since understaffed clinical environments have no capacity absorb additional manual steps. ^[21] reinforced this through 200 hours of observation and 21 nurse interviews in a tertiary hospital, finding that the most consistently cited barrier to nursing CDSS adoption was poor fit between the system and existing task demands, not the underlying algorithm. Any re-triage or alerting system must therefore generate output passively rather than requiring active nurse interaction to function.

Constraint 3: Competing demands on porter and transfer staff. ^[7] identified that porters were frequently pulled between transferring admitted patients to wards and transporting patients to investigations, creating conflict over which task should take priority and consequently delaying both. This is a downstream constraint rather than a triage-stage constraint, but it illustrates that flow-level AI interventions at the front of the department cannot resolve outflow bottlenecks further down the pathway; the digital triage tool addresses input and throughput but does not eliminate output-stage delays, a limitation consistent with the three-component ED crowding model used by ^[19], who note that input, throughput, and output factors interact and that solving one does not automatically resolve the others.

Additional implementation evidence

Beyond the three named constraints, two further studies strengthen the case for the specific design choices proposed in this project. ^[20] conducted a human-factors workflow integration analysis of a clinical decision support tool deployed directly in an emergency department, drawing on 134 barrier and facilitator excerpts from 12 ED physicians, and found that adoption depended heavily on whether the tool fit into the existing timing, flow, and patient journey scope of clinical work rather than requiring a separate interaction step, directly supporting the design requirement in Constraint 2 that the system surface passively rather than add interaction steps. ^[15] extended this finding in a post-implementation usability evaluation of the same tool, identifying that despite high perceived usability in simulation, real-world clinical use remained low, with poor workflow integration cited as the dominant barrier, reinforcing the rationale for AI Plug-in 5's plain-language, NEWS2-aligned explanation surfaced directly within the existing clinician review step rather than as a separate system.

Stakeholders

Stakeholder	Top concern
Triage nurse	The system must not add steps to an already understaffed, interruption-heavy workflow
ED consultant	Alerts must be clinically actionable, not noise; alert fatigue research shows that systems generating high false-positive rates are quickly ignored or overridden by staff ^[11]
Registration clerk	Free-text chief complaint quality directly affects NLP plug-in accuracy
Clinical IT Lead	Full audit trail, data governance compliance, zero real patient data in pilot
Patient and family advocate	Assurance that automated scoring does not delay care for patients who present calmly but deteriorate quietly

Risk Name	Category	Likelihood	Impact	Mitigation	Signal of Success
Urban-rural distribution mismatch	Equity	High	High	Research coordinator runs stratified performance analysis by age, sex and presenting complaint at weeks 4, 8 and 12; results reviewed before any Ministry recommendation	Subgroup performance parity within 5% across demographic groups at each review point
Covariate shift at district-level deployment	AI-Technical	High	High	Restrict deployment claims to Mercer General for full 12-week window; re-validate at minimum two additional Caribbean sites before rollout	Model AUROC remains above 0.80 at each new deployment site
Automation bias toward NEWS2 output	Ethical	Medium	High	Shadow-mode pilot keeps nurses unaware of system outputs for full 12 weeks; final decision remains with clinician at all times	Manual triage override rate remains above 95% throughout pilot, verified by weekly case log
Re-triage alert fatigue	Operational	Medium	High	Re-triage clock set to 30-minute minimum; alerts suppressed for already-escalated patients; override rate tracked weekly	Alert override rate remains below 20%; any week exceeding threshold triggers immediate review
NLP misclassification of Creole-influenced language	AI-Technical	Medium	High	Outputs below 0.70 confidence threshold suppressed and replaced with manual review flag; threshold reviewed weekly by Clinical IT Lead	NLP false-positive rate below 10% on weekly audit; all suppressed outputs reviewed within 24 hours
Missing patient consent disclosure	Ethical	Medium	Medium	Patient advocate drafts plain-language disclosure approved by ED Board before pilot day one; notice displayed at registration in	Consent notice confirmed displayed before day one and at each weekly site check

				English and Spanish	
Pseudonymised data re-identification	Ethical	Low	High	No real identifiers stored; all data pseudonymised at point of capture; Clinical IT Lead signs off governance plan before pilot	Zero reportable data incidents during 12-week pilot
Silent deterioration under-triage	AI-Technical	Medium	High	NLP flags high-risk language in chief complaint when vitals appear stable; trajectory tracking catches gradual deterioration across consecutive readings	Zero retrospectively identified under-triage cases in weekly ED consultant review
Manual triage deskilling	Operational	Low	Medium	Shadow-mode prevents clinicians acting on AI output; competency assessed by blinded nurse test at week 0 and week 12	Mean manual triage accuracy does not decline more than 5% between assessments
Subgroup degradation in patients over 75	Equity	Medium	High	Stratify pilot analysis by under 65 and over 75 age groups; flag atypical vital sign presentations for additional ED consultant review	AUROC difference between age groups remains below 0.05 across all review points

Risks

Risk 1: Training-Distribution Mismatch Between Mercer General and Underserved Caribbean Populations

The most serious equity risk to this project is that a model trained exclusively on data from a single urban tertiary emergency department may systematically underperform for patient groups underrepresented in that training data. ^[6] demonstrated through a scoping review of clinical AI papers published in PubMed that models trained on relatively homogeneous datasets and populations poorly representative of underlying diversity limit generalisability and risk producing biased AI-based decisions, finding that the vast majority of clinical AI research originates from a small number of high-income countries whose patient demographic

profiles differ substantially from Caribbean populations. This risk is directly applicable to the Mercer General context, where the patient demographic profile, disease burden and healthcare utilisation patterns diverge significantly from the North American and European populations that dominate the AI triage literature.

The most consequential documented example of this failure class is provided by ^[14], whose landmark study demonstrated that a widely used commercial health algorithm deployed across United States hospitals systematically assigned Black patients lower risk scores than equally sick White patients because it used healthcare cost as a proxy for health need. The study found that only 17.7% of patients the algorithm assigned for additional care were Black; correcting the bias would have raised that figure to 46.5%. The root mechanism was that less money is spent on Black patients for the same conditions due to structural access barriers, causing the algorithm to falsely conclude that Black patients were healthier. ^[14] concluded that omitting a race variable from a model does not guarantee that the algorithm will not produce racially biased outputs, because bias enters through proxy variables that correlate with race without naming it. This finding directly informs the equity audit requirement built into this project's risk register, where stratified performance analysis by age, sex and presenting complaint is conducted at weeks 4, 8 and 12 of the pilot before any Ministry recommendation is made.

Risk 2: Alert Fatigue from High-Frequency Re-Triage Prompts

The automated 30-minute re-triage clock in Component 2 introduces a risk that prompts issued too frequently or when the clinical situation has not changed will generate alert fatigue, causing clinicians to suppress or ignore notifications including those signalling genuine deterioration. ^[1] identified alert fatigue as a recurring implementation failure in AI-assisted emergency decision support, arguing that systems generating high false-positive rates are rapidly ignored or overridden by staff, undermining the clinical value of the tool entirely and creating a situation where real deterioration signals are lost within noise.

^[3] reinforced this finding through an analysis of 429 safety reports submitted to the FDA Manufacturer and User Facility Device Experience (MAUDE) database for AI and machine learning-enabled medical devices, identifying alert management failures as a recurring theme across reported safety incidents and concluding that an AI patient safety programme is urgently needed to monitor alert behaviour in deployed clinical systems. The mitigation built into this project's design is two-fold: the re-triage prompt is suppressed for patients who have already been escalated to a higher priority category, and pilot data will track alert override rates weekly, with any week exceeding 20% overrides triggering an immediate review of prompt frequency settings before the following week's operations.

Risk 3: Automation Bias Inducing Passive Acceptance of AI Risk Output

Automation bias describes the tendency of clinicians to defer to automated system outputs without independent verification, particularly under high workload and time pressure conditions that are characteristic of a busy Caribbean emergency department. ^[5] found in a systematic review of 76 papers on clinical decision support systems that the primary reason for inappropriate alert behaviour in live hospital practice is that the vast majority of decision support systems were not built using established human factors engineering methods, creating tools that nudge clinicians toward passive acceptance rather than active clinical judgement. This risk is particularly acute in the context of a system that maps outputs onto NEWS2 language that staff already trust, since that familiarity may reduce critical scrutiny of individual score outputs.

^[17] demonstrated in a systematic review that XAI explanations can actively undermine clinical trust or cause clinicians to accept flawed outputs when the explanations are contradictory or poorly designed, confirming that the design of interpretability mechanisms is as consequential as their presence. The shadow-mode design of this pilot directly mitigates automation bias by keeping nurses unaware of system outputs for the full 12-week window, ensuring that manual triage competency is not compromised during the validation period and that the system cannot be used as a substitute for clinical judgement before its safety and accuracy have been independently verified.

Real-World AI Harm Case Study

The Optum Racial Bias Algorithm (2019)

Source: Obermeyer Z, Powers B, Vogeli C, Mullainathan S. Dissecting racial bias in an algorithm used to manage the health of populations. *Science*. 2019;366(6464):447-453. doi:10.1126/science.aax2342

1. What Happened

In 2019 a landmark peer-reviewed study led by Ziad Obermeyer revealed a critical patient safety failure in a widely used healthcare algorithm developed by Optum, part of UnitedHealth Group, deployed across hospitals in the United States to identify patients who should be enrolled in high-risk care management programmes. The algorithm used predicted future healthcare spending as a proxy for health need. Because Black patients historically incurred lower healthcare costs due to unequal access, structural barriers and differences in care utilisation, the algorithm systematically assigned them lower risk scores than equally sick White patients. ^[17] found that only 17.7% of patients the algorithm assigned to receive extra care were Black; correcting the bias would have raised that figure to 46.5%. An estimated 200 million patients annually were processed through similar algorithms across the

United States, meaning the harm operated silently at scale for years before external researchers identified it.

2. Why It Happened

The root cause was a flawed problem formulation at the design stage, not a technical execution error. Developers chose healthcare spending as the target variable because cost data was available, measurable and historically predictive. They did not account for the structural inequity embedded in that variable: Black patients spend less on healthcare for the same conditions because they access care less frequently due to systemic barriers including financial cost, geographic access and historical institutional distrust. The algorithm performed exactly as designed. It accurately predicted costs. But predicting costs was not equivalent to identifying health need, and that conflation was never challenged during development or validation. ^[6] identified this class of failure as pervasive across clinical AI, noting that models trained on utilisation data systematically underrepresent populations with limited access to care because the learning signal itself encodes the inequity.

3. What the System Failed to Anticipate

The system failed to anticipate two distinct things. First, it failed to anticipate that a proxy variable can encode a protected characteristic even when that characteristic is explicitly excluded from the model. Omitting race from the feature set did not prevent the algorithm from producing racially biased outputs because the proxy variable of healthcare cost correlated strongly with race through structural pathways the model could not observe or correct for. Second, the system failed to anticipate that standard aggregate validation metrics would pass even while subgroup performance was severely and inequitably degraded. The algorithm achieved strong overall predictive accuracy for cost prediction while simultaneously producing deeply inequitable allocations across racial groups, a discrepancy that aggregate accuracy metrics are not designed to detect. ^[3], in their analysis of 429 FDA MAUDE safety reports for AI-enabled medical devices, identified the absence of disaggregated subgroup validation as a recurring structural precursor to safety incidents across clinical AI deployments, confirming that overall performance metrics are insufficient as a safety standard.

4. What Would Have Caught It

Three mechanisms would have caught this failure before deployment at scale. First, a mandatory pre-deployment equity audit requiring the developer to report model performance stratified by race, sex and socioeconomic status would have surfaced the subgroup disparity before any hospital adopted the tool. Second, replacing healthcare cost with a direct measure of disease burden, such as comorbidity count or validated clinical severity score, would have removed the racially confounded proxy from the target variable entirely at the design stage. Third, the failure was ultimately identified by independent academic researchers with no

financial interest in the product, suggesting that mandatory third-party algorithmic auditing before deployment at scale would have caught the problem years earlier and prevented harm to millions of patients. ^[1] argued specifically that mitigating algorithmic bias in emergency decision-making requires equity audits conducted by parties external to the development team as a condition of deployment approval, not as an optional post-market activity.

Technical vs Human-Factors Failure

The technical failure was the use of a structurally biased proxy variable without disaggregated subgroup validation during development. The human-factors failure was the absence of institutional challenge to the problem formulation at the design stage and the absence of a clinical governance mechanism at the deploying hospitals that would have required equity performance verification before adoption. Neither failure alone caused the harm at the scale it occurred. The combination of a technically valid but ethically unexamined model and an institutional environment that trusted aggregate accuracy metrics without interrogating subgroup outcomes produced a system that caused inequitable harm across millions of patients for years before external scrutiny identified it.

Relevance to the Mercer General Pilot

This case directly informs three design decisions in this project. The shadow-mode pilot prevents the system from making autonomous care allocation decisions, removing the immediate harm pathway that the Optum algorithm created. The equity audit built into the risk register requires stratified subgroup performance analysis by age, sex and presenting complaint before any deployment recommendation is made to the Saint Cedric Ministry of Health. The Glass-Box Logic Layer ensures every risk output is traceable to a specific vital sign or nurse note, giving clinicians the interpretability needed to recognise and override a result that appears clinically inconsistent with the patient in front of them. ^[17] concluded that the choice of convenient, seemingly effective proxies for ground truth can be an important source of algorithmic bias in many contexts, a lesson this project addresses at the design stage rather than leaving to post-market discovery.

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